Preprint

SUPPLEMENTARY MATERIAL

Supplementary Material: The Tetrad Scheme and the Two Bounded Regions

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Abstract

The main letter derives the stellated octahedron from the half-space inequalities of four derived tetrahedral directions (Route 1). This note provides a second, independent route to the same result (Route 2). The defining equation of the tetrad, $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu$, is quadratic in the tetrad, and it forces $(\det e)^2 = \det g / \det \eta$, an equation with two roots. Its solution space (scheme) has exactly two connected components, one over each root, related by spatial parity. Each component carries a regular tetrahedron with a $3 + 1$ face decomposition. The compound of the two components is the stellated octahedron. The metric determines only $(\det e)^2$, so it cannot distinguish the components. This argument uses no half-space bounding, no recession cones, and no sign-combination enumeration.

Keywords: tetrad, scheme, algebraic geometry, Lorentz group, spatial parity, root-blindness, stellated octahedron, proof

The Tetrad Equation Is Quadratic

The tetrad $e^a{}_\mu$ is defined by the relation

$$g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu\, e^b{}_\nu, \qquad \text{or} \qquad g = e^T \eta\, e, \tag{1}$$

which is quadratic in e. If $e^a{}_\mu$ is a solution, then so is $\Lambda^a{}_b\, e^b{}_\mu$ for any $\Lambda \in O(1,3)$, because $O(1,3)$ preserves η_{ab} by definition. Conversely, every solution is of this form. Taking determinants in Eq. (1) gives $(\det e)^2 \det \eta = \det g \neq 0$, so every solution is invertible; for two solutions e and e', the matrix $\Lambda = e' e^{-1}$ satisfies $\Lambda^T \eta \Lambda = e^{-T} g\, e^{-1} = \eta$. The full solution space at any point is therefore the orbit of a single tetrad under the Lorentz group $O(1,3)$, and distinct Λ give distinct solutions.

This solution space is what algebraic geometry calls the *scheme* of the equation [1, 2]: the set of all solutions, considered as a geometric object in its own right, without selecting a preferred root. Precisely, the sixteen entries of e are the unknowns, Eq. (1) is a set of quadratic equations in them, and

$$X_g = \operatorname{Spec} \mathbb{R}[e^a{}_\mu] \,/\, (e^T \eta\, e - g)$$

is the affine scheme they define. Its real points are the tetrads, and by the argument above X_g is isomorphic to the orthogonal group scheme $O(\eta)$, through $\Lambda \mapsto \Lambda e_0$ for any one tetrad e_0.

The Scheme Has Two Components

Taking determinants in Eq. (1) once more,

$$(\det e)^2 = \frac{\det g}{\det \eta} \equiv c > 0, \tag{2}$$

where $c > 0$ because g and η have the same signature. This is an equation of the form $y^2 = c$, as $x^2 = a$ is for a single unknown. Every tetrad has $\det e = +\sqrt{c}$ or $\det e = -\sqrt{c}$, never both. The scheme therefore splits as $X_g = X_+ \sqcup X_-$, the parts over the two roots, each open and closed. Neither splits further: under $X_g \cong O(\eta)$ the two parts are $SO(\eta)$ and its coset, and $SO(\eta)$ is connected even over $\mathbb{C}$, where it is $SO(4, \mathbb{C})$ [3]. The scheme has exactly two connected components, the two roots of Eq. (2).

The Lorentz group $O(1,3)$ has four connected components [4], distinguished by two discrete operations:

- spatial parity P: $e^0 \to e^0$, $e^i \to -e^i$ $(i = 1, 2, 3)$,
- time reversal T: $e^0 \to -e^0$, $e^i \to e^i$.

The four components are $SO^+(1,3)$, $P \cdot SO^+(1,3)$, $T \cdot SO^+(1,3)$, and $PT \cdot SO^+(1,3)$. A change of frame $e \to \Lambda e$ multiplies $\det e$ by $\det \Lambda$, so it exchanges the roots of Eq. (2) if and only if $\det \Lambda = -1$. P and T each have determinant -1 and exchange them; $PT = -I$ has determinant $+1$ and does not. The four real components are the two roots, each split by the direction of the time leg.

The main letter's construction uses only the spatial legs of the tetrad: the four derived directions v_1, v_2, v_3, v_4 are linear combinations of e^1, e^2, e^3. Time reversal T acts only on e^0 and leaves the spatial legs unchanged. To compare the frames of one observer, hold the time leg fixed and future-pointing. The changes of frame that do so are $\Lambda = \operatorname{diag}(1, Q)$ with $Q \in O(3)$, and $\det \Lambda = \det Q$: for these frames the root of Eq. (2) is the orientation of the spatial legs.

The spatial scheme thus has exactly two distinct components:

- the identity component $SO(3)$, which rotates the four derived directions rigidly,
- the parity component $P \cdot SO(3)$, which negates all three spatial legs.

The continuous part, $SO(3)$, rotates the four derived directions rigidly and produces no new geometry. The discrete part, P, sends every $v_i \to -v_i$, producing a second set of four unit vectors with the same pairwise inner product $c = -1/3$. Four unit vectors in $\mathbb{R}^3$ with common pairwise inner product $-1/3$ form the vertices of a regular tetrahedron [6]; their convex hull is a bounded solid. The identity component gives TetA. The parity component gives TetB. Precisely, P maps TetA $= \{\max_i v_i \le 1\}$ to $\{\max_i(-v_i) \le 1\}$, the all-flipped sign combination, which is the second bounded region of Route 1.

Three facts make this identification exact. First, P is the canonical member of its component: the only elements of $O(3)$ commuting with every rotation and reflection are $\pm I$, since commuting with all diagonal sign matrices and all permutation matrices forces a multiple of I, hence $\pm I$, and of these only $-I$ has determinant -1. Second, parity is not a rotation in disguise: the four directions span $\mathbb{R}^3$, so $Qv_i = -v_i$ for all i forces $Q = -I$, and no element of $SO(3)$ carries the frame of TetA to that of TetB leg by leg. Third, this rests on the spatial dimension being odd: $\det(-I) = (-1)^D$, and in an even number of spatial dimensions $-I$ is a rotation, so parity would produce no new component.

Two cautions keep the statement exact. As unlabelled point sets the tetrahedra are congruent, since a regular tetrahedron is its own mirror image (a 90° rotation about one axis maps the vertices of TetA onto those of TetB), so the components differ in the order of the legs and the region of space they occupy, not in shape. And reversing every leg, $e \to -e = PT\,e$, keeps $\det e$ but also reverses the time leg; its spatial legs are those of Pe. The identification of root with tetrahedron therefore holds for future-pointing frames, the setting of the main letter.

Both Components Are Necessarily Present

The scheme of a polynomial equation contains all its roots by definition. The quadratic structure of the tetrad equation and the two components of its scheme together guarantee that both TetA and TetB exist in the solution space. Both satisfy the same defining equation with the same metric, and the metric cannot tell them apart: by Eq. (2) it determines $(\det e)^2$ and never $\det e$, and every quantity built from g, including the Christoffel symbols, the curvature, the Einstein tensor and the Einstein–Hilbert action, takes the same value on both components. Retaining one component and discarding the other is consistent, but no equation written in the metric can make that selection.

The textbook convention of restricting local frames to the proper orthochronous Lorentz group $SO^+(1,3)$ [4, 5] corresponds to working within a single connected component. This is a valid computational choice, but it is a restriction of the full scheme to one component. The scheme itself contains both. The choice between the signature conventions $(-,+,+,+)$ and $(+,-,-,-)$ is a different matter: it changes the sign of η as a whole, leaves the Lorentz group unchanged, and selects no component.

Each Component Carries a $3+1$ Decomposition

In the main letter's coordinates, whose axes are the frame's spatial legs, the four face normals of TetA, expressed as sign patterns on the three spatial axes, are:

$$(+,+,+), \quad (+,-,-), \quad (-,+,-), \quad (-,-,+). \tag{3}$$

One face normal has uniform sign: $(+,+,+)$, the direction $e_1 + e_2 + e_3$. Three face normals have mixed signs. This is the $3+1$ partition of the four faces that the main letter reads as one temporal and three spatial. The partition is fixed by the frame's spatial legs: under the tetrahedron's own symmetries all four faces are equivalent, so it is a property of the tetrahedron relative to the frame, not of the tetrahedron alone.

Under spatial parity P, the face normals of TetB are:

$$(-,-,-), \quad (-,+,+), \quad (+,-,+), \quad (+,+,-). \tag{4}$$

Again, one face normal has uniform sign: $(-,-,-)$. Three have mixed signs. TetB carries its own $3+1$ decomposition, relative to the same legs.

The stellated octahedron therefore carries the $3+1$ pattern in both tetrahedra, with the uniform-sign normals $(+,+,+)$ and $(-,-,-)$ exchanged by spatial parity, independently in both bounded regions.

Summary

The main letter derives the stellated octahedron by classifying the sixteen half-space sign combinations and proving exactly two yield bounded regions (Route 1). The present note derives the same result from the quadratic structure of the tetrad equation (Route 2):

	Route 1 (letter)	Route 2 (this note)
Input	half-space inequalities	tetrad eqn. $g = \eta\, e\, e$
Method	recession cone test	scheme of quad. eqn.
Key fact	$\sum v_i = 0$ (normals sum to zero)	$(\det e)^2 = c$ has two roots
Output	2 of 16 bounded	2 components of the scheme

Both routes arrive at the same stellated octahedron with the same $3 + 1$ face decomposition in each tetrahedron. Route 2 uses no half-space construction and Route 1 no quadratic structure; both take the four tetrahedral directions from the main letter's signature-to-frame supplement. A rejection of one leaves the other intact. The loop between equation and geometry is closed: the two roots of the tetrad equation are the two bounded regions, TetA and TetB, exchanged by spatial parity. The metric is blind to which root a frame occupies, and keeping one is a convention.

References

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